

Francisco D. Serna

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Education

University of Minnesota - Twin Cities

Bachelor of Engineering in Mechanical Engineering

Expected May 2028 | Minneapolis, MN

GPA: 3.46/4.0

Internship Experience

Abbott Laboratories | *Electrophysiology Research and Development Intern*

May 2026 - Aug. 2026 | Plymouth, MN

- Led an evaluation study to validate polyurethane stopcock tubing as a replacement for existing PVC tubing in an introducing catheter by assembling test samples and executing **Design Verification (DV)** methods, including aging, water/air leak, tensile strength, and bend/kink radius testing.
- Collaborated in the development of a new test method using **Gage R&R** to characterize the variability of nitinol baskets through torque testing and evaluations of lateral stability
- Evaluated shaft inner diameter (ID) design changes through friction testing, development of **Design Control Models (DCMs)**, and **tolerance stack-ups** to assess the impact of increased ID on shaft interactions within the handle assembly using **MiniTab**, **SolidWorks**, and **Excel**

Cretex Medical - rms Company | *Manufacturing Engineering Intern*

May 2025 - Dec. 2025 | Coon Rapids, MN

- Created a visual database based on research on the different types of tumbling media, cycle times, and their impact on post-tumbling properties, helping operators achieve a desired finish, saving up to 15 minutes between tumbling cycles
- Modeled and designed fixtures specific to individual components for laser marking and EDM wiring in **SolidWorks**
- Conducted inspections and calibrations of component iterations and machines weekly

Work Experience

Lifetime Fitness Inc. | *Facility Operations Supervisor*

Jan. 2023 - Present | Coon Rapids, MN

- Directed club upkeep by overseeing 48 team members, ensuring the practice of standard club procedures
- Exceeded standard customer service values with existing and new members to ensure a friendly, clean environment
- Ensured team flow by completing scheduling, task lists, area designations, and other administrative tasks.

Projects & Skills

IOT Smart Trash Can

- Designed and developed an IoT-enabled smart trash can using a Particle Photon 2, integrating ultrasonic sensing, load cell weight measurement, and servo-actuated mechanisms in **C++**, with real-time status indication via cloud
- Implemented cloud-connected monitoring and control through the Particle Cloud API, enabling remote operation and automated actuation, and modeled a 3D-printed enclosure in **SolidWorks**

Autonomous Spider-Bot

- Designed and built a 12-servo hexapod-style robot in C++ on an Arduino Uno, engineering a custom pin/slider mechanism for coordinated leg rotation that lets the robot grip and pull itself across a surface
- Modeled the full mechanical assembly in **Creo** and **SolidWorks** and **3D-printed** all structural parts, integrating an IR proximity sensor to trigger an autonomous 90° turn when obstructions come within 25 cm

Technical Skills:

- SolidWorks, C++, Minitab, MicroVu, PTC Windchill Workgroup Manager, PTC Creo, PTC Creo View, ANSI Drawing Standards, Sheetmetal design, Microsoft 365, Google Workspace, Bilingual in Spanish

College Activities/Societies:

- President's Emerging Scholars Program (PES)
- Peer Mentor for first- and second-year students in PES
- American Society of Mechanical Engineers (ASME)
- Teaching Assistant for EE 1301: Intro to C/C++